

Ram Ravi

+1 (267) 968-4052 | raviramgopal@gmail.com | ramravi2403.github.io

SUMMARY

Machine Learning Engineer with an M.S. in Data Science and 8+ years building high-throughput, scalable data systems across Morgan Stanley, Athena Capital Research, and Exelon. More recently, applied work spans end-to-end ML pipelines, NLP/RAG, computer vision, and reinforcement learning, with a track record of collaborating across research and engineering teams to take systems from concept to production.

EDUCATION

Drexel University

Philadelphia, PA

Master of Science in Data Science, GPA: 4.0/4.0

June 2025

- Post-Baccalaureate Certificate in Computational Artificial Intelligence & Machine Learning

The Hong Kong University of Science and Technology

Hong Kong

Bachelor of Engineering in Electronic Engineering

June 2014

- Dean's List (Spring 2013, 2014)

TECHNICAL SKILLS

Languages: Python, C#, Java, C, SQL, R

Libraries: Pandas, NumPy, Scikit-Learn, PyTorch, PySpark, Matplotlib, LangChain, YOLO, OpenCV, PIL, Optuna

Analytics & ML: Machine Learning, Deep Learning, Time Series, LLM, Data Cleaning and Visualization, Statistics

Cloud & MLOps: AWS (RDS, Elastic Beanstalk), Docker, MLflow

Databases & Developer Tools: PostgreSQL, Git, Mercurial, Unix, Roboflow

EXPERIENCE

FutureMap LLC

Oakland, CA

AIML Engineer

Jul 2026 – Present

- Building an end-to-end data and ML pipeline using **Python, PostgreSQL, AWS, and LLMs** to ingest, match, enrich, classify, and summarize organizational and financial data
- Developed LLM-based classification and summarization workflows and deployed an initial application using **AWS RDS and Elastic Beanstalk**

Drexel University

Philadelphia, PA

Machine Learning Researcher (Advisor: Prof. Shahin Jabbari)

Sep 2025 – Jul 2026

- Investigated robust algorithmic recourse and model-extraction risk using **ROAR, Optimal L_∞ recourse**, and **DualCF** under constrained query budgets
- Built a reproducible end-to-end ML experimentation pipeline spanning target-model training, counterfactual generation, surrogate extraction, and evaluation across **HELOC, GMSC, and SBA** datasets, achieving up to **0.98+ extracted AUC** on tabular models
- Evaluated more than **2,600 configurations each** for HELOC and GMSC, finding that higher robustness budgets could weaken DualCF extraction agreement, particularly for Optimal L_∞ , while ROAR remained more stable on deep targets

Exelon (Baltimore Gas & Electric)

Baltimore, MD

Advanced Analysis & Modeling (Co-op)

Jul 2024 – Jan 2025

- Built scalable, high-throughput Python data pipelines to automate forecasting and capacity planning workflows for Distributed Energy Resources (DER) across **1,600+ feeders**, reducing analysis time from **~1 week to under 10 minutes**
- Built distributed grid topology and connectivity models from LoadSEER hierarchy - reconstructing circuit component ordering (reclosers, transformers, fuses) to enable accurate load tracking, prevent overloading, and support outage reconfiguration across BGE's distribution network
- Developed geospatial data pipelines to resolve incomplete DER installation addresses and infer GPS coordinates, improving data quality for downstream systems (IntellioConnect, LoadSEER)

Athena Capital Research

Senior Trade-Operations Engineer

Hong Kong

Sep 2018 – Sep 2023

- Collaborated cross-functionally with quantitative researchers, developers and engineers to support model development and evaluation pipelines - conducting post-trade analytics in Python to identify execution inefficiencies
- Analysed market data across multiple exchanges to surface price return correlations and trading inefficiencies - delivering findings to quantitative researchers as inputs to model refinement
- Developed real-time monitoring tools in Python to detect and flag trading-system errors - reducing investigation and response time from up to **2 hours to under 10 minutes**
- Managed trading risk limits and capital allocations across strategies - ensuring compliance with exchange requirements

Morgan Stanley

Technology Associate / Analyst

New York & Hong Kong

Aug 2014 – Apr 2018

- Built scalable data and automation pipelines supporting risk and trading systems, improving system reliability and reducing manual effort by **10+ hours/week**
- Led migration of Hong Kong's risk account database to Sybase ASE 16.0 - ensuring long-term support and system stability
- Designed and presented a continuous system monitoring framework to senior management, improving production anomaly detection coverage across Institutional Equity systems

PROJECTS

Bookshelf Scanner | YOLOv8, Google Cloud Vision, LangChain, Python

2025

- Built an end-to-end ML pipeline for book discovery integrating object detection, OCR, LLM-based metadata extraction, and embedding-based retrieval
- Trained and evaluated multiple **YOLOv8** variants on a ~20,000-image custom dataset; best model (**YOLOv8x-ORB**) achieved **mAP50 of 93.7%** (precision 86.4%, recall 90.9%)
- Benchmarked OCR engines (**Google Cloud Vision**, Tesseract, Tesseract) using Jaccard and edit-distance metrics; selected Google Cloud Vision (**Jaccard 0.79, ~2x improvement**)
- Designed semantic retrieval system using **Sentence-BERT** embeddings and LLM-generated metadata to enable similarity-based book recommendations

Bipedal Locomotion via Deep RL | PyTorch, Optuna, Gymnasium

2025

- Implemented and tuned **A2C with GAE** from scratch in PyTorch for continuous control, and benchmarked against **PPO and SAC** in Gymnasium's BipedalWalker-v3 environment
- Developed a modular **hyperparameter tuning framework using Optuna** (Bayesian optimization + pruning), running 30+ trials over 13+ hours to optimize key parameters
- Achieved **135.20 mean reward (A2C)** and **321.58 (SAC)** across 50 evaluation trials under a unified training setup
- Analyzed policy generalization by transferring a hardcore-trained SAC agent to the standard environment, achieving a mean reward of **248.02**

Resume Scanner | LangChain, OpenAI

2024

- Built an end-to-end **RAG** pipeline using **LangChain** and **OpenAI** to match resumes to job descriptions and identify personalized skill gaps
- Designed evaluation metrics (embedding similarity, keyword overlap) to assess recommendation quality and improve alignment between resumes and job requirements
- Extended the system to recommend relevant learning resources (e.g., Coursera courses) to help users address identified skill gaps

PUBLICATIONS

Y. Zhao, **R. Ravi**, et al. "PATE: Property, Amenities, Traffic & Emotions Coming Together for Real Estate Price Prediction" *2022 IEEE 9th Intl. Conf. on Data Science and Advanced Analytics (DSAA)*

Y. Zhao, S. Shi, **R. Ravi**, et al. "H4M: Heterogeneous, Multi-source, Multi-modal, Multi-view, Multi-distributional Dataset for Socioeconomic Analytics in Beijing" *2022 IEEE 9th Intl. Conf. on Data Science and Advanced Analytics (DSAA)*

PERSONAL

CFA Level 1, PADI Advanced Open Water Diver, Drexel Finance and Investment Group, HKUST International Students Soccer Team